MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**November, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**C**
Course Name	**Building Materials**	**Course Code**	**CIE 2123**
Semester	**III**	**Max Marks**	**50**
Date		**Duration**	**3hrs**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**
- **Use of IS 10262 2019and IS 456 2000 is allowed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A.	Comment on large scale use of Industrial wastes as coarse aggregates.	**2**	**1**	**4**	3
1B.	Elaborate the significance of sieve analysis for coarse aggregates for its use in concrete.	**3**	**1**	**3**	4
1C.	Describe the bogue's compounds and comment on their rate of hydration and heat release.	**5**	**2**	**2**	**4**
2A.	List out the precautions to be taken when the concrete is to be transported a long way before it is placed?	**2**	**3**	**2**	**2**
2B.	Comment on microstructure development in concrete with and without micro fillers.	**3**	**2**	**4**	**4**
2C.	Explain the mechanism of supplementary cementitious materials in enhancing properties of concrete	**5**	**3**	**2**	**2**
3A.	Design a concrete mix for M30 concrete and 100mm slump. Binder paste consists of Portland cement with specific gravity of 3.15 and class F flyash with specific gravity of 2.35. Coarse aggregates obtained from nearby quarry is of sp gravity 2.67 while a mixture of quarry dust and crushed rock is used as fine aggregates with specific gravity of 2.66 and falls into zone II.	**8**	**2**	**3**	**2**
3B.	Sketch the load spread in a stretcher bond.	**2**	**4**	**3**	**3**
4A.	Elaborate is the benefit of using flyash blocks as masonry units in comparison with clay bricks	**2**	**4**	**3**	**3**
4B.	Sketch the various components of a typical floor system.	**3**	**4**	**4**	**3**

4C.	What are the possible alternative masonry blcoks used for construction. Elaborate steps that can be taken to avoid brittle failure in masonry.	**5**	**5**	**3**	**2**
5A.	Elaborate the two essential components of floors with figure.	**2**	**5**	**4**	**3**
5B.	Differentiate between single and double Flemish bond.	3	**5**	**4**	**3**
5C.	Compare plastic paint and enamel paint.	**5**	**5**	**2**	**3**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO	Total Marks
1	***4***
2	***9***
3	***13***
4	***13***
5	***11***

BL	Total Marks
2	**12**
3	**30**
4	**8**
5	